

The advisory a farmer never gets.

AgriMitra Climate reads a specific field — its weather, its satellite vegetation, its soil, its crop — and returns one decision for the week ahead, with every assumption shown.

86%

of Indian farmers hold under 2 hectares

1:1000

extension officers to farmers

15–25%

of yield lost to pests and disease

₹1.4L cr

upper bound of that loss, annually

THE USER

Ravi farms 0.8 hectares in Alwar.

He knows rain is coming. What he does not know is whether, given his loamy soil, his limited irrigation, the state of his wheat this week and the forecast, he should irrigate now or wait.

- The nearest extension officer covers roughly a thousand farms like his.
- Weather apps give a forecast, not a decision.
- A disease scanner names a disease, then stops.
- Advice that arrives on WhatsApp is generic to the district, not to the field.

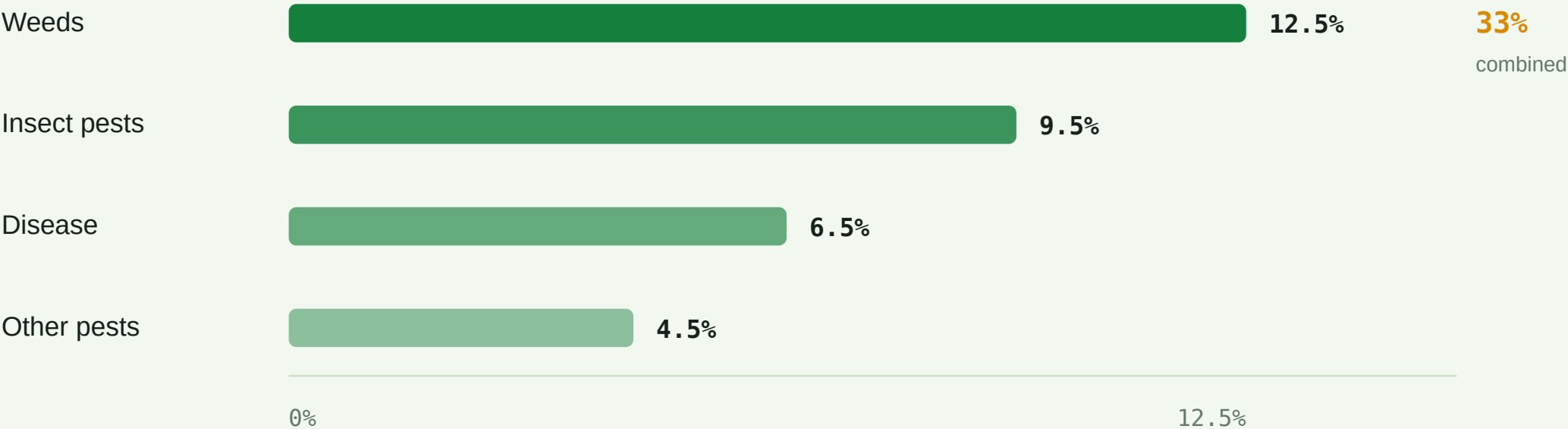
Rain is coming. So — do I irrigate today, or lose the window?

Farmers who receive early, accurate guidance cut crop losses by **30–60%**. The signal exists. It does not reach the field in a form anyone can act on.

PROBLEM • MAGNITUDE

A third of the crop is lost before it leaves the field.

Share of Indian production lost, by cause. Every one of these is visible in a leaf or a soil-moisture reading days before it becomes a loss.



Shares are of total production loss attributed to pests in Indian agriculture.

Six apps, no answer.

The data a farmer needs already exists. It arrives in fragments, none of which knows about the others, and none of which knows the field.

Weather is not advice

A 7-day forecast is an input. Turning it into “irrigate Thursday, not today” needs the soil, the crop stage and the water available.

Diagnosis stops at the name

A scanner returns “Early blight” and ends. It never reaches the risk score, and never changes what the farmer is told to do next.

Satellite stays with agronomists

NDVI is routine in enterprise platforms and absent from the farmer's phone.

Advice is district-wide

Two farms in one district with different soil and irrigation get identical guidance.

Confidence is never shown

Tools present estimates as measurements. A farmer cannot tell which number to trust.

States cannot share models

Every state rebuilds its own crop dataset and risk logic. Nothing is interoperable.

THE SOLUTION

One field. Five signals. One decision.

Deterministic logic computes the score. Gemini explains it. The model never invents a measurement — an architectural rule, enforced in code.

- 01 **Field profile** — location, crop, season, soil, irrigation. Entered once; GPS fills district and state.
- 02 **Live weather** — current conditions and a 7-day forecast for that coordinate.
- 03 **Satellite vegetation** — Sentinel-2 NDVI reduced over the field via Google Earth Engine.
- 04 **Leaf scan** — a photograph routed to MobileNetV2, escalating to Gemini multimodal below the confidence threshold.
- 05 **Risk score** — five weighted factors, and a written note for every factor that had no evidence.
- 06 **Advisory** — Gemini narrates the computed numbers in English, हिन्दी, ગુજરાતી or తెలుగు.

RISK ENGINE

**Water 0.30 · Heat 0.20 · Disease
0.20 · Rainfall 0.15 · Vegetation 0.15**

When the weather service is unreachable, heat and rainfall are **dropped from the weighted score** and the remaining weights renormalised — never silently read as zero.

SHOWN TO THE FARMER

“No leaf analysis on record; disease risk uses the neutral baseline (30/100). Scan a leaf to replace it.”

WHAT MAKES IT DIFFERENT

It tells the farmer what it does not know.

Agricultural advice moves money and food. A confident wrong answer is worse than an admitted gap — so provenance is a first-class field in the API, not a footnote in the UI.

— Every measurement carries

`is_live` ,

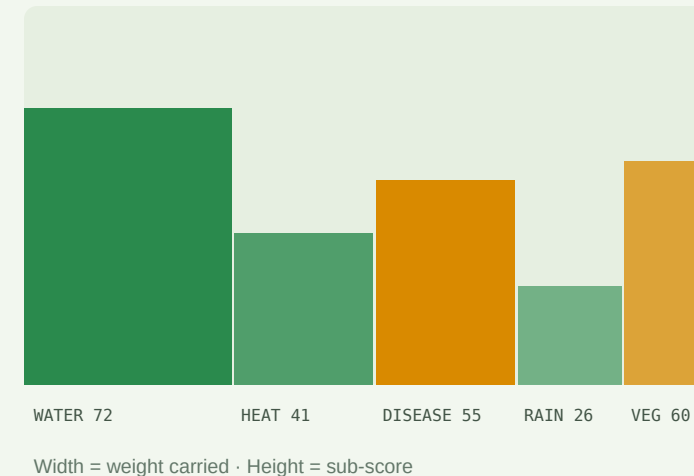
`degraded` and

`is_demo` — the types make them non-optional, so a screen cannot render a number without the means to label it.

— Demo satellite data says

“Demo dataset — not a live satellite observation.”

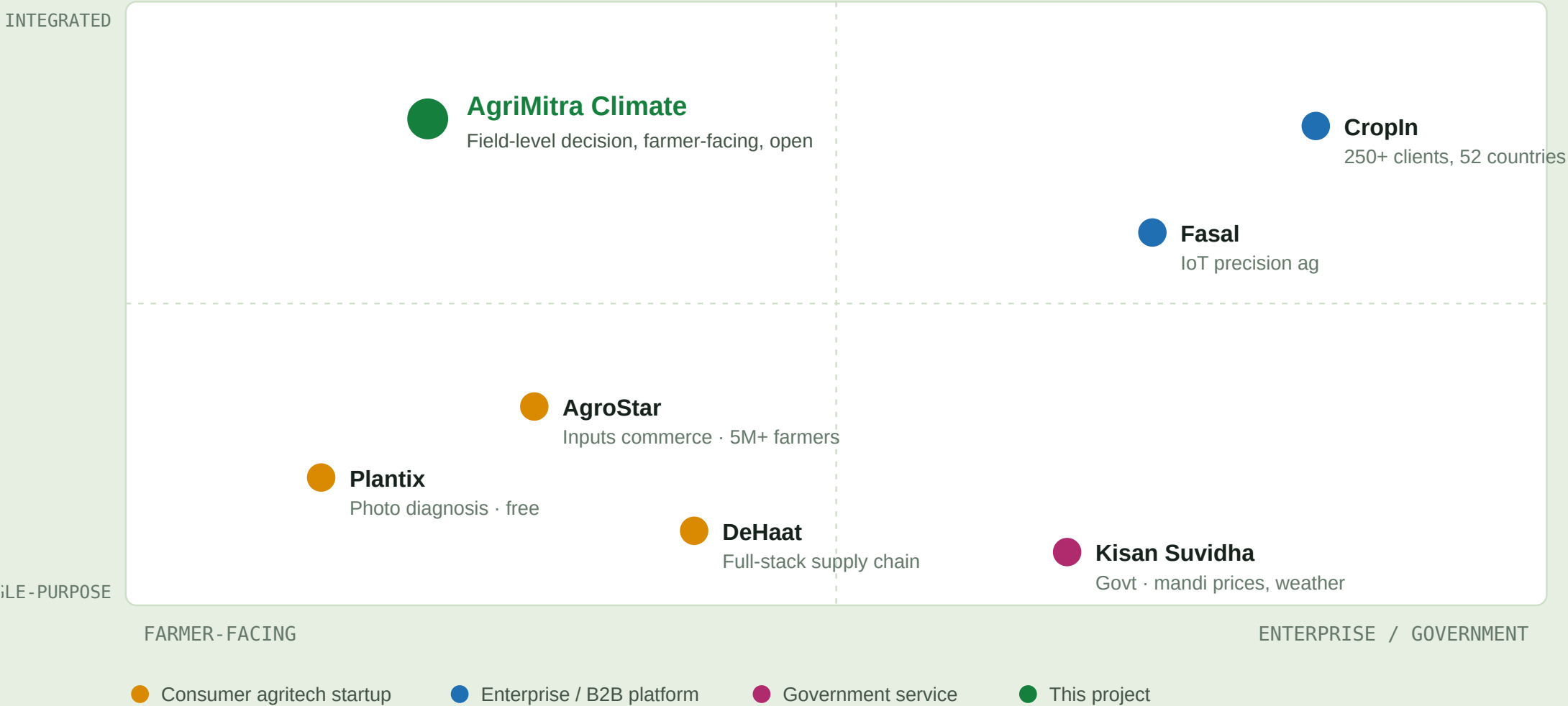
- With no Gemini key, the advisory is labelled rule-based rather than passed off as model output.
- Mandi prices return a 503 when unconfigured. Never an estimate.
- The leaf scanner names the 14 crops it was trained on, and warns before you photograph a crop outside them.



The signature view. It answers “what is driving this number” — which five independent progress bars cannot. A factor with no evidence is absent, not drawn at zero.

Everyone owns one layer. Nobody owns the decision.

Horizontal: who the product serves. Vertical: whether it answers a single question or produces a field-level decision.



Placement reflects each product's primary positioning, not its full feature list.

The landscape, honestly.

These are strong products. The gap is not that they are bad — it is that none of them closes the loop from satellite and soil to a decision the farmer can act on, and none is built to be shared between states.

PRODUCT	TYPE	CORE STRENGTH	SCALE	GAP THIS LEAVES
Plantix	STARTUP	Best-in-class photo disease diagnosis, free, many Indian languages	—	Diagnosis only; no soil, satellite or field risk
AgroStar	STARTUP	Content-led input commerce; farmers post crop photos for advice	5M+ farmers	Advisory serves product sales; not a neutral decision layer
DeHaat	STARTUP	Full-stack inputs, advisory and market linkage across 12 states	\$700M+ val.	Supply-chain first; satellite risk is not the product
CropIn	ENTERPRISE	Satellite + ML farm intelligence; Gemini partnership since 2024	52 countries	Sold to agribusiness, not reachable by a 0.8 ha farmer
Fasal	ENTERPRISE	IoT sensors driving hyper-local irrigation and spray timing	—	Requires hardware per farm; capital cost excludes smallholders
Kisan Suvidha	GOVERNMENT	Official mandi prices, weather and advisories, free	National	Reference data, not analysis; nothing is field-specific
AgriMitra Climate	THIS PROJECT	Satellite + soil + weather + leaf → one explainable field risk score	Hackathon build	Unproven at scale — see roadmap

Indian agritech raised **\$241M across 60 rounds in 2025**; the market is projected to grow from **\$974M (2025) to \$2.52B by 2034**.

ARCHITECTURE

Google AI, load-bearing.

Each service below has a caller and a user-reachable code path. Where credentials are absent the feature degrades to a labelled fallback — it is never faked to pad a list.

REASONING

Gemini API · Vertex AI

Advisory, assistant, district interventions, crop-ranking explanations. Structured output, temperature 0.2, and a system instruction forbidding invented measurements. Vertex gives a second auth path using service-account IAM, so Cloud Run holds no API key.

VISION

Gemini multimodal

Second opinion whenever the local classifier falls below 0.70 — which is every crop outside its 14 trained ones, i.e. most Indian staples.

GEOSPATIAL

Earth Engine · Maps

Sentinel-2 NDVI reduced over a 150 m buffer, cloud-filtered to 30%. Maps Platform renders the district risk layer.

LANGUAGE

Speech & Translation

Cloud Speech-to-Text and Text-to-Speech for the browsers with no Web Speech API — Android WebView, most in-app browsers. Translation localises the scheme directory.

DATA

BigQuery · Cloud Storage

Every computed risk is appended to a warehouse time series, including which factors carried evidence. Leaf photos are retained as the evidence behind a diagnosis.

PLATFORM

Cloud Run · Secret Manager

Three services, non-root images, health checks. Secrets injected at runtime. Pub/Sub fans out NDVI refresh; Redis caches weather and satellite reads.

Public data: `data.gov.in` Agmarknet mandi prices · Open-Meteo forecast and historical archive · OpenStreetMap reverse geocoding · Sentinel-2 via Earth Engine.

THE THESIS

A crop dataset is a public good. Ship it as one.

The challenge asks for an interoperable network that lets Indian states share agricultural data models. Two things in this system already *are* portable models — they simply need publishing as such.

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The crop dataset

— ten crops with season, soil, water, temperature and rainfall envelopes.

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The risk model

— five named weights, four bands, and the rules for dropping a factor with no evidence.

Published as versioned JSON Schema behind a federation API, Rajasthan's crop model becomes something Maharashtra can adopt, critique or replace — without a bilateral integration.

FEDERATION, PRIVACY FIRST

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Aggregates only.

No farm record, no farmer record, no coordinate finer than a district centroid crosses a boundary.

—

A k-anonymity floor.

A district with fewer than ten farms is omitted — “1 farm, CRITICAL, Wheat” identifies a person.

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Signed manifests.

A peer verifies before ingesting, so a hostile node cannot poison another state's dashboard.

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Enforced by test

, not by policy prose.

Open licence · open standard · documented · no PII · platform independent — the five Digital Public Good criteria, each mapped to where the code satisfies it.

Working software, not slideware.

26

API endpoints, OpenAPI-
documented

90

automated tests across two
services

10

screens, 4 languages, light and
dark

100

seeded farms across 10
districts

Running end to end

- Farm registration with GPS → district resolution → risk in one flow, against MongoDB Atlas.
- Live Open-Meteo weather and measured seasonal rainfall from the historical archive.
- Leaf scan: 20 ms rejection of non-leaves, 160–260 ms inference, 612 ms for a 3600 px photo.
- District roll-up aggregating 100 farms into risk bands and dominant crops.

Bugs found and fixed by testing, not reading

- A confident *healthy* leaf produced a disease risk of 95/100 .
- The leaf detector rejected every diseased colour — rust, blight, necrosis — as “not a plant”.
- Crop rainfall was compared in mm/hour against a mm/season requirement, so that criterion never scored.
- A dead weather API was read as “0 °C, no rain” instead of as missing.

ROADMAP

What comes next, in order.

Ranked by value to a farmer or an officer, divided by effort.

NEXT	WHAT IT UNLOCKS	EFFORT
Federation layer	Published schema plus peer endpoints — the interoperable network the challenge asks for	1 day
FAO crop coefficients	Replaces prototype constants with citable FAO-56 values	2 hours
Regenerative scoring	Rotation-aware ranking: a legume after a cereal scores above the cereal repeated	0.5 day
Vertex AI AutoML	Next-season risk forecast trained on the BigQuery history, shown beside the computed score, never replacing it	1 day
Indian crop disease model	Wheat, rice, mustard, chickpea — the crops the current checkpoint has never seen	Ongoing
ISRO Bhuvan · IMD	Indian sovereign satellite and meteorological sources behind the existing provider interface	1 day

Give the field a voice it can act on.

Not another weather app. Not another disease scanner. The first layer that reads a specific field across every signal available — and is honest about the ones it is missing.

For the farmer

A weekly decision, in their language, with the working shown.

For the officer

District risk hotspots and AI intervention priorities, from real aggregation.

For the state

A crop model and a risk model any other state can adopt, critique or replace.

Sources — Agriculture Census 2015–16 (86.1% of holdings under 2 ha; marginal holdings average 0.38 ha) · ICAR / CEEW on crop losses (15–25% of yield; ₹0.9–1.4 lakh crore annually; weeds 12.5%, insects 9.5%, disease 6.5%, other 4.5%) · extension coverage ≈ 1 officer per 1,000 farmers; early accurate guidance reduces losses 30–60% · Tracxn, Inc42 and AgroSpectrum India on funding and market size (\$241M across 60 rounds in 2025; \$974M in 2025 to \$2.52B by 2034) · company profiles from Tracxn, TechCrunch and Decentro. Figures describe the Indian market as reported by those sources; placements on slide 07 are our reading of each product's primary positioning.